TABLE OF CONTENTS

SR No	Contents	Page No
	Abstract: Brief overview of the project objective, method, and outcomes	**01**
	Acknowledgements: Thanks to guides, faculty, peers, or institutions	**02**
1	**Introduction**	**03**
	Background of the Problem	
	Aim of the Mini Project	
	Relevance to Mechanical Engineering	
2	**Literature Survey / Review**	**05**
	Summary of Related Works in Literature review	
	Technologies or Methods Used in Similar Projects	
	Research Gap or Motivation	
3	**Project Planning and Methodology**	**07**
	Planning / Gantt Chart (if used)	
	Tools, Materials, and Software Used	
	Block Diagram or Process Flow (if applicable)	
4	**Design and Implementation**	**10**
	Concept Design	
	Sketches / CAD Drawings	
	Basic Calculations (if any)	
	Fabrication or Simulation Process	
5	**Results and Discussion**	**13**
	Output or Prototype Demonstration	
	Challenges Faced	
	Interpretation of Results	
6	**Conclusion**	**15**
	Key Achievements	
	What was Learned	
	Possible Improvements or Extensions	
7	**References**	**16**
	Books, Journals, Websites, Article	

Abstract

The Piezoelectric Sidewalk project aims to generate electrical energy from the mechanical pressure exerted by pedestrians while walking. Piezoelectric discs installed beneath movable tiles convert this pressure into electrical energy, which is rectified into DC and stored for later use. The generated energy can power low-power devices such as LED indicators and pathway sensors. This project demonstrates an innovative and sustainable energy-harvesting method for smart infrastructure, although the output depends on applied force, pedestrian traffic, and system efficiency.

Acknowledgements

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We are thankful to our department and institution for providing the necessary facilities, laboratory resources, and academic support required to complete this project.

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1. INTRODUCTION

1.1 Background of the Problem:

The world's energy requirements are continuously increasing due to population growth, industrial development, urbanization, and the increasing use of electronic devices. Conventional energy sources such as coal, petroleum, and natural gas are limited and contribute significantly to environmental pollution and greenhouse gas emissions. Therefore, there is a growing need to investigate renewable and alternative energy sources.

Human movement represents a large amount of mechanical energy that is normally wasted. Every time a person walks, mechanical energy is transferred from the person's body to the ground. A portion of this energy can potentially be captured using suitable energy-harvesting technologies.

Piezoelectric materials have the ability to generate electrical charges when mechanical stress or pressure is applied to them. This phenomenon is known as the **piezoelectric effect**.

A piezoelectric sidewalk uses this principle by placing piezoelectric elements beneath a walking surface. When pedestrians walk over the surface, their weight applies force to the piezoelectric elements. The resulting electrical output can be collected and stored. Thus, a piezoelectric sidewalk provides an interesting approach to converting otherwise wasted mechanical energy from human footsteps into useful electrical energy.

1.2 Aim of the Mini Project

The main aim of this project is: **To design and develop a prototype piezoelectric sidewalk capable of converting mechanical energy generated by human footsteps into electrical energy.**

Objectives

The major objectives are:

1. To understand the piezoelectric effect.

2. To study the conversion of mechanical energy into electrical energy.
3. To design a suitable sidewalk structure containing piezoelectric elements.
4. To fabricate a small-scale prototype.
5. To collect and measure the generated electrical output.
6. To store the generated energy using suitable electronic components.
7. To demonstrate the harvested energy using a low-power load such as an LED.
8. To investigate possible applications of piezoelectric energy harvesting.

1.3 Relevance to Mechanical Engineering: The project is directly related to mechanical engineering because it involves the application of mechanical force, structural design, material selection, stress, deformation, and energy conversion.

The sidewalk structure must be designed to:

- withstand repeated human loads,
- transfer pressure efficiently to the piezoelectric elements,
- protect the electrical components,
- provide adequate mechanical strength,
- minimize deformation of the walking surface,
- maintain durability under repeated loading.

The project also introduces concepts such as:

- mechanical loading,
- stress and strain,
- force transmission,
- material properties,
- structural design,
- energy conversion,
- product development.

Therefore, the project combines **mechanical design and electrical energy harvesting** to develop a sustainable engineering solution.

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2. LITERATURE SURVEY / REVIEW

2.1 Summary of Related Works

Piezoelectric energy harvesting has been studied as a method of recovering small amounts of energy from mechanical vibrations, pressure, movement, and human activity.

Researchers have investigated the use of piezoelectric materials in:

- roads,
- sidewalks,
- railway platforms,
- stairs,
- floors,
- shoes,
- bridges,
- industrial machinery,
- vehicle systems.

The basic concept is to place piezoelectric materials at locations where mechanical forces or vibrations are frequently produced.

Human footsteps are particularly interesting because public locations can experience thousands of footsteps every day. Although the energy obtained from an individual step is relatively small, repeated footsteps can provide a useful source of low-power energy.

2.2 Technologies or Methods Used in Similar Projects

Different methods can be used to harvest energy from human movement.

1. Piezoelectric Energy Harvesting

Piezoelectric discs or plates generate electrical charges when mechanical stress is applied.

2. Electromagnetic Energy Harvesting

Mechanical motion is used to move a magnet relative to a coil, producing electricity through electromagnetic induction.

3. Triboelectric Energy Harvesting

Electrical charges are generated through contact and separation between different materials.

4. Hydraulic Energy Harvesting

Foot pressure can be used to drive a hydraulic mechanism connected to an energy conversion system.

For this project, **piezoelectric energy harvesting** was selected because it is relatively simple, compact, lightweight, and suitable for demonstrating energy conversion on a small scale.

2.3 Research Gap or Motivation

Although piezoelectric energy harvesting has been studied extensively, several challenges remain when applying it to practical sidewalks.

Some important challenges include:

- low electrical output from individual piezoelectric elements,
- durability under repeated loading,
- efficient force transfer,
- protection of piezoelectric elements,
- energy storage,
- cost of large-scale implementation,
- variation in output due to different pedestrian weights and walking patterns.

The motivation of this project is therefore to develop a **simple, low-cost prototype** that demonstrates the feasibility of harvesting energy from human footsteps.

3. PROJECT PLANNING AND METHODOLOGY

3.1 Planning / Gantt Chart

The project can be planned as follows:

Week	Activity
Week 1	Selection of project topic
Week 2	Literature survey
Week 3	Concept development
Week 4	Component selection
Week 5	Mechanical design
Week 6	Electrical circuit design
Week 7	Prototype fabrication
Week 8	Circuit assembly
Week 9	Testing and measurements
Week 10	Result analysis
Week 11	Improvements
Week 11	Final documentation

3.2 Tools, Materials, and Software Used

Hardware Components

Sr. No.	Component	Purpose
1	Piezoelectric discs	Convert mechanical pressure into electrical energy
2	Wooden/acrylic platform	Walking surface
3	Springs	Provide controlled movement and force distribution
4	Wires	Electrical connections
5	Bridge rectifier	Converts AC/pulsating output into DC
6	Capacitor	Stores electrical energy

7	LED	Demonstrates generated energy
8	Resistors	Circuit protection/current limiting
9	Diodes	Prevent unwanted current flow
10	Rechargeable battery/cell	Optional energy storage
11	Connecting board/PCB	Circuit assembly

Software

Depending on the project implementation, the following software may be used:

- Arduino IDE

3.3 Block Diagram / Process Flow

The basic working process is:

Human Footstep

↓

Mechanical Pressure

↓

Piezoelectric Elements

↓

Electrical Charge Generation

↓

Rectifier Circuit

↓

Capacitor / Battery

↓

DC Output

↓

LED / Sensor / Low-Power Device

Working Principle

When a person steps on the sidewalk, the top plate moves slightly downward. This movement applies mechanical force to the piezoelectric elements.

The piezoelectric elements convert the applied mechanical stress into electrical energy.

The electrical output from multiple piezoelectric elements can then be connected to a rectifier circuit. The rectified output is stored in a capacitor or rechargeable battery.

The stored energy can subsequently be used to operate a low-power electrical load.

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4. DESIGN AND IMPLEMENTATION

4.1 Concept Design

The proposed piezoelectric sidewalk consists of a rigid base, piezoelectric elements, a movable top plate, and a mechanism for transferring foot pressure to the piezoelectric elements.

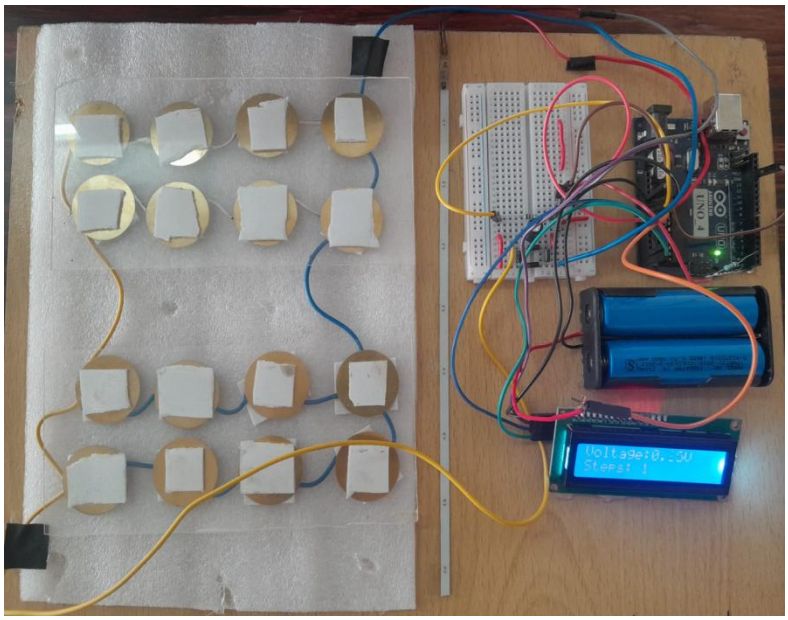
A simplified structure consists of:

1. Top walking plate
2. Piezoelectric discs
3. Supporting springs
4. Bottom/base plate
5. Electrical wiring
6. Rectifier and storage circuit

When a person steps on the top plate, the plate moves slightly downward. This movement creates mechanical stress in the piezoelectric elements.

The arrangement should be designed so that the force is distributed rather than concentrated at a single point.

4.2 Prototype



4.3 Fabrication or Simulation Process

The fabrication process consists of the following steps:

Step 1: Design the Platform

A suitable top plate and base plate are designed according to the required dimensions.

Step 2: Prepare the Piezoelectric Arrangement

Multiple piezoelectric discs are positioned underneath the top plate.

Step 3: Install Springs

Springs can be placed between the top and bottom plates to provide controlled movement and return the top plate to its original position.

Step 4: Connect the Piezoelectric Elements

The piezoelectric elements are electrically connected according to the required circuit configuration.

Step 5: Install the Rectifier

The output is connected to a bridge rectifier to obtain a suitable DC output.

Step 6: Add Energy Storage

A capacitor or rechargeable battery can be connected to store the harvested energy.

Step 7: Connect the Load

An LED or other low-power device is connected to demonstrate the electrical output.

Step 8: Test the Prototype

The prototype is tested by applying repeated footsteps and measuring voltage and current.

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5. RESULTS AND DISCUSSION

5.1 Output or Prototype Demonstration

The fabricated prototype was tested by applying pressure to the sidewalk surface through repeated footsteps.

The piezoelectric elements generated electrical voltage whenever mechanical pressure was applied.

The output voltage depends on several factors, including:

- applied force,
- number of piezoelectric elements,
- arrangement of the elements,
- speed of loading,
- deformation,
- electrical connection,
- mechanical structure.

A low-power LED can be used as a simple demonstration of the generated electrical energy.

Observation Table

No. of Footsteps	Measured Voltage (V)
1	0.04
2	0.17
3	1.02
4	1.92
5	2.45

5.2 Challenges Faced

During the development of the piezoelectric sidewalk, several challenges may be encountered.

1. Low Electrical Output

Individual piezoelectric discs generally produce limited useful energy. Multiple elements may therefore be required.

2. Mechanical Loading

The piezoelectric elements must be protected from excessive mechanical force because excessive loading can damage them.

3. Force Distribution

Uneven pressure can cause some piezoelectric elements to receive more force than others.

4. Electrical Losses

The energy generated by the piezoelectric elements may be reduced because of losses in wires, rectifiers, and storage components.

5. Structural Stability

The top platform must be strong enough to support users while still allowing sufficient movement to transfer force to the piezoelectric elements.

5.3 Interpretation of Results

The prototype demonstrates that mechanical energy from human footsteps can be converted into electrical energy using piezoelectric materials.

The output increases with repeated mechanical excitation because each footstep produces another electrical pulse.

However, the amount of energy produced by an individual step is relatively small. Therefore, piezoelectric sidewalks are more suitable for low-power applications rather than replacing conventional electrical power systems.

Potential applications include:

- LED lighting,
- pedestrian counters,
- wireless sensors,
- display indicators,
- IoT sensors,
- environmental monitoring devices,
- low-power communication systems.

The project demonstrates that energy that would otherwise be lost during walking can be partially recovered and utilized.

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6. CONCLUSION

The Piezoelectric Sidewalk project demonstrates the conversion of mechanical energy from human footsteps into electrical energy using piezoelectric elements. The generated energy can be stored and used for low-power applications such as LEDs and sensors. The project highlights the potential of piezoelectric technology for sustainable energy harvesting.

6.1 Key Achievements

1. Studied the piezoelectric effect.
2. Designed a basic sidewalk structure.
3. Integrated piezoelectric elements.
4. Developed an energy collection circuit.
5. Obtained electrical output from footsteps.
6. Explored low-power applications.

6.2 What Was Learned

- Piezoelectric effect and energy conversion.
- Mechanical design and material selection.
- Rectification and energy storage.
- Prototype fabrication, testing, and teamwork.

6.3 Possible Improvements or Extensions

1. Increase the number of piezoelectric elements.
2. Improve the mechanical structure and circuit efficiency.
3. Use batteries or supercapacitors for energy storage.
4. Integrate IoT and microcontrollers to monitor footsteps and electrical output.
5. Implement the system in public places such as railway stations, colleges, and shopping malls.

7. REFERENCES

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